

E.3 Radioactive decay

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **26 marks**

Show your working.

Question 1 — The banana bunker

(7 marks)

Bananas are mildly radioactive (potassium-40). A worried customer builds an elaborate lead-lined fruit bowl.

- (a) Rank alpha, beta and gamma radiation by *ionizing* power, and separately by *penetrating* power. [2]
- (b) State the relationship between the two rankings, and explain physically why the most ionizing radiation is the least penetrating. [3]
- (c) State the material typically sufficient to stop each of the three radiations. [2]

Question 2 — Balancing the books

(7 marks)

Potassium-40 ($^{40}_{19}\text{K}$) can decay by beta-minus emission.

- (a) Write the complete decay equation, including the daughter nucleus (calcium, Ca) and all emitted particles. [3]
- (b) State the two conservation rules your equation must satisfy, and verify each with the numbers. [2]
- (c) Describe what happens inside the nucleus during beta-minus decay, at the level of nucleons. [2]

Question 3 — The rival samples

(6 marks)

Sample X starts with $4N_0$ nuclei of a nuclide with half-life T . Sample Y starts with N_0 nuclei of a nuclide with half-life $2T$.

- (a) Calculate the number of nuclei remaining in each sample after a time $2T$. [3]
- (b) Hence write the ratio X : Y at that moment. [1]
- (c) Explain why tracking only the starting amounts, or only the half-lives, both give wrong answers here. [2]

Question 4 — The impatient technician

(6 marks)

A detector near a small source reads 120 counts per minute. With the source removed entirely, the same detector reads 20 counts per minute.

- (a) Name what the 20 counts per minute measures, and give two of its origins. [2]
- (b) Predict the detector reading after one half-life of the source. (Careful: it is not 60.) [3]
- (c) State why background must be subtracted *before* any half-life analysis, not after. [1]

Answers

Question 1 — The banana bunker

- (a) Ionizing power: $\alpha > \beta > \gamma$. Penetrating power: $\gamma > \beta > \alpha$.
- (b) The two rankings are exact reverses of each other, and that is no coincidence. Being strongly ionizing means giving energy away at almost every encounter with the atoms in its path. Radiation that does so spends its energy within a very short distance and therefore cannot penetrate far. Penetration is precisely the *result* of interacting weakly: gamma rays travel a long way because they so rarely interact at all.
- (c) Alpha: a sheet of paper, or a few centimetres of air. Beta: a few millimetres of aluminium. Gamma: many centimetres of lead or metres of concrete, and even then it is attenuated rather than completely stopped.

Question 2 — Balancing the books

- (a) ${}^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{Ca} + {}^0_{-1}\text{e} + \bar{\nu}_e$ (an antineutrino accompanies every beta-minus emission).
- (b) Nucleon number A : $40 = 40 + 0$. Proton number, and hence charge, Z : $19 = 20 + (-1)$. Both balance.
- (c) Inside the nucleus a *neutron* converts into a *proton*, emitting an electron and an antineutrino: $n \rightarrow p + e^- + \bar{\nu}_e$. The electron is created at that moment; it was not sitting in the nucleus beforehand.

Question 3 — The rival samples

- (a) Sample X: half-life T , so in a time $2T$ it undergoes *two* halvings, leaving $\frac{4N_0}{2^2} = N_0$. Sample Y: half-life $2T$, so in a time $2T$ it undergoes only *one* halving, leaving $\frac{N_0}{2}$.
- (b) $X : Y = N_0 : \frac{1}{2}N_0 = 2 : 1$.
- (c) Looking only at the starting amounts suggests a fixed 4 : 1 forever, which ignores that X is halving twice as often. Looking only at the half-lives suggests Y must win because it decays more slowly, which ignores that X started with four times as much. The number remaining depends on *both* the initial quantity and how many halvings fit into the elapsed time.

Question 4 — The impatient technician

- (a) It measures the *background* count rate. Any two origins from: cosmic rays; radon gas in the air; naturally radioactive rocks, soil and building materials; traces from medical or industrial sources.
- (b) Subtract the background first. The source alone contributes $120 - 20 = 100$ counts per minute. After one half-life the source contributes 50, and the background is still there, so the detector reads $50 + 20 = 70$ counts per minute. (Halving the raw 120 to get 60 is the trap: it wrongly halves the background too.)
- (c) Because the background does not decay: it is a constant offset, not part of the exponential. Leaving it in distorts every ratio, as $120 \rightarrow 70$ is plainly not a halving, so any half-life extracted from the raw readings would be wrong.